

## 1. What is React?

React is a JavaScript library developed by Facebook to build fast and interactive user interfaces using reusable components. It follows a component-based architecture.

## 2. Why React is Used?

React helps in building scalable applications. It improves performance using Virtual DOM and allows code reusability through components.

## 3. How React Works?

React uses a Virtual DOM. When state changes, React creates a new Virtual DOM, compares it with the previous one (diffing), and updates only the changed parts in the real DOM.

## 4. React Workflow

index.html loads first → main.jsx runs → App component renders → child components render → UI is displayed.

## 5. Components

Components are reusable pieces of UI. Types: Functional components (modern) and Class components (older).

## 6. JSX

JSX is a syntax that looks like HTML but is actually JavaScript. It gets converted into `React.createElement` using Babel.

## 7. Props

Props are used to pass data from parent to child components. They are read-only.

## 8. State

State is used to manage dynamic data inside a component. It can change over time and re-renders the UI.

## 9. Hooks

Hooks allow you to use state and lifecycle features in functional components. Example: `useState`, `useEffect`.

## **10. Event Handling**

React handles events using camelCase syntax like onClick, onChange, and uses functions to handle them.

## **11. Conditional Rendering**

React allows rendering UI based on conditions using if-else, ternary operator, or logical AND.

## **12. Lists and Keys**

Lists are rendered using map(). Keys help React identify elements uniquely for efficient updates.

## **13. Forms**

React forms are controlled components where form data is handled using state.

## **14. API Calls**

APIs are called using fetch or axios inside useEffect hook.

## **15. Routing**

React Router is used to navigate between pages without reloading the page.

## **16. Advantages of React**

Fast performance, reusable components, strong community, easy to learn, and widely used in industry.

## **17. Limitations of React**

Only handles UI, needs additional libraries for full functionality, and frequent updates can be confusing.